FREE COHOMOLOGY MAKES THE ETA IDEALS PRINCIPAL, GENERATED BY POWERS OF THE NONZERODIVISOR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The counted proof span contains the author's sentence 'In this case it is clear that I_i(M^bullet, f) = (f^{r_i}) is principal.'; that assessment is delivered verbatim and not expanded. The three references inside the counted proof span (including the Derived Categories reference) are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Base change of the ideals I_i(M^bullet, f) along localization at a prime. invoked at proof lines 28036-28037. Reduces the statement to the local case.
- Well-definedness of I_i(M^bullet, f) up to quasi-isomorphism. invoked at proof lines 28037-28038. Permits replacing M^bullet by a convenient quasi-isomorphic complex.
- A bounded complex of free modules with free cohomology over a local ring splits as the direct sum of its cohomology modules. invoked at proof lines 28039-28043. Allows assuming that all differentials vanish.

Source reference. Original lemma 0GM4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20744–20744. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 28013: sources/stacks-more-algebra/pinned-original-source.tex, source lines 28013–28013. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 27893: sources/stacks-more-algebra/pinned-original-source.tex, source lines 27893–27893. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let A be a ring, let $\mathfrak{p} \subset A$ be a prime ideal, and let $f \in A$ be a nonzerodivisor. Let $M^\bullet$ be a bounded complex of finite free A-modules. If $H^i(M^\bullet)_\mathfrak{p}$ is free for all i, then $I_i(M^\bullet, f)_\mathfrak{p}$ is a principal ideal and in fact generated by a power of f for all i.*

Proof. We may assume A is local with maximal ideal $\mathfrak{p}$ by Lemma 28013. We may also replace $M^\bullet$ with a quasi-isomorphic complex by Lemma 27893. By our assumption on the freeness of cohomology modules we see that $M^\bullet$ is quasi-isomorphic to the complex whose term in degree i is $H^i(M^\bullet)$ with vanishing differentials, see for example Derived Categories, Lemma 0GM4. In other words, we may assume the differentials in the complex $M^\bullet$ are all zero. In this case it is clear that $I_i(M^\bullet, f) = (f^{r_i})$ is principal. □